Callum Huntington

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Education

PhD in Mathematics and its Applications 2023–2026
University of Naples Federico II, Italy
thesis: *On compositions of graphs, signed graphs, and gain graphs*
supervisor: Francesco Belardo

MSc in Mathematics (with Distinction) 2020–2021
University of Sheffield, United Kingdom
thesis: *Hyperbolic geometry and sandpiles*
supervisor: Paul Johnson

BSc in Mathematics (with Class One Honours) 2017–2020
University of Sheffield, United Kingdom

Research Experience

PhD Researcher 2023–2026
University of Naples Federico II, Italy

Research focus: spectral graph theory, i.e. what the eigenvalues of a graph's matrices reveal about the graph itself. Most of my work concerns signed and complex unit gain graphs, where every edge carries a sign or a complex number of modulus one, with emphasis on structural operations (joins, products, corona constructions) and switching equivalence. Developed spectral methods using linear algebra and group representations for gain graphs over arbitrary groups.

Collaborations with researchers at Niccolò Cusano University, Italy (S. Spessato, A. Donno, M. Cavaleri), University of Belgrade, Serbia (Z. Stanić), University of the Western Cape, South Africa (O. Waka), IIT Bhubaneswar, India (S. Barik), Federal University of Rio Grande do Sul, Brazil (E. Oliveira, V. Trevisan), and Eindhoven University of Technology, The Netherlands (A. Abiad, A. Khramova).

Publications

Published and forthcoming

C. Huntington, O. Waka. Spectra of gain *Q* and *R* graphs via *G*-phases and group representations. *Ars Combinatoria*, to appear.

C. Huntington, F. Belardo, S. Barik. Graphs with integer reciprocal eigenvalue properties. *Special Matrices*, to appear.

- **C. Huntington**. Spectra and balance of signed graphs obtained from vertex splittings. *Discussiones Mathematicae Graph Theory*, vol. 46, pp. 987–1015, 2026. DOI: 10.7151/dmgt.2643.

- **C. Huntington**, S. Spessato, F. Belardo. On H-joins of complex unit gain graphs and their stability. *Discrete Mathematics*, vol. 349, no. 7, art. 115069, 2026. DOI: 10.1016/j.disc.2026.115069.

- **C. Huntington**. The H-join operation of signed graphs constrained by indexing maps. *RAIRO-Operations Research*, vol. 59, no. 5, pp. 2501–2515, 2025. DOI: 10.1051/ro/2025100.

- F. Belardo, **C. Huntington**. Signed graphs with strong (anti-)reciprocal eigenvalue property. *Special Matrices*, vol. 12, no. 1, art. 20240017, 2024. DOI: 10.1515/spma-2024-0017.

Under review and in preparation

- **C. Huntington**, Z. Stanić, F. Belardo. Further spectral relations between oriented graphs and corresponding signed graphs. *Under review*.

- F. Belardo, A. Donno, **C. Huntington**, S. Spessato. A switching-stable corona product of signed graphs. *Under review*.

- F. Belardo, **C. Huntington**, E. Oliveira, V. Trevisan. Intervals of limit points for B_α-spectral radii. *Under review*.

- A. Abiad, **C. Huntington**, A.P. Khramova. A switching method for obtaining cospectral signed and gain sun graphs. *In preparation*.

- F. Belardo, **C. Huntington**, S. Spessato. Regularity and strong regularity of gain graphs over the fourth roots of unity. *In preparation*.

- **C. Huntington**. Products derived from gain subdivision graphs via group representations. *In preparation*.

- **C. Huntington**. Counterexamples to a conjecture on Hamiltonian cycles in signed graphs. *In preparation*.

Conferences

2nd Joint Meeting Brazil–Italy in Mathematics September 2026
Messina, Italy
invited talk: Graphs and signed graphs with integer reciprocal eigenvalue properties

Real-world Applications of Geometry and Algebra 2026 June 2026
Eindhoven, The Netherlands

Combinatorics 2026 May 2026
Naples, Italy

Mini Workshop of Topology and Algebra 2025 December 2025
Cape Town, South Africa
invited talk: The H-join of complex unit gain graphs and their stability

Workshop on Spectral Graph Theory 2025 October 2025
Gramado, Brazil
contributed talk: Spectral relations between oriented graphs and corresponding signed graphs

12th PhD Summer School in Discrete Mathematics September 2025
Koper, Slovenia
contributed talk: The H-join of complex unit gain graphs and their stability

CIME Summer School: Modern Perspectives in Approximation Theory July 2025
Cetraro, Italy

IRGTADS Seminars June 2025
Online
invited talk: The H-join operation on signed graphs

Group Theory in Salerno March 2025
Salerno, Italy

1st Chinese–South-East European Conference on Discrete Mathematics and Applications June 2024
Belgrade, Serbia
contributed talk: Constructions of nonregular cospectral signed graphs

University of Pisa Summer School: Introduction to Machine Learning in Geosciences July 2022
Pisa, Italy

Other Skills and Qualifications

Languages: English (native), Italian (B1).

Programming: Python and SageMath (used throughout research for computational experiments); familiar with R and MATLAB.

Refereeing activity: Served as a peer reviewer for several international journals in graph theory and discrete mathematics.